Roadmap

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Objectives of programming languages

- easy to learn
- easy to get programs right
- execute fast
- safe (avoid disaster)

The course objectives

- get how different programming languages approach these goals differently
- topics
  - types
  - code reusability (generics, subtyping, inheritance, etc.)
  - memory management/safety
  - performance
  - building compilers

The course objectives

- practices / exercises: you choose two languages from below and do course work in them
  - Go
  - Julia
  - OCaml
  - Rust

The course format

- lecture + exercises (during class time and home work)
- exercise environment : Jupyter
- initially work individually
- after a few weeks, we group students
- each group will be two or three students, each working on two different languages
- we discuss approaches to the above objectives taken by different languages, within and across groups
- you are expected to engage in these discussions and other activities (not just to listen to talks and get things done)

Evaluation

- small coding-centric assignments in Jupyter (a few times)
- reflective essay (every week, due the end of the next day)
- participations (esp. in discussions)
- a final report (building a simple compiler by default; other options are possible)
- some Jupyter exercises are for self learning (no need to submit)
- *assignments / reports subject for evaluation/grading are indicated as assignments in UTOL*
- no exams

Reflective essay

- every week, you write a short reflective essay that expresses such things as
  - what *you* have learned (conceptualize/internalize experiences)
  - what came through *your* mind while listening to the talk and working on assignments
  - how *you* worked on the exercise (where you struggled, how you got help, how useful was AI, etc.)

Today

- answer a survey on your programming language experiences and the preferred (natural) language
- play with the Jupyter environment
  - work on a language or two you choose
  - write a few programs in them
- practice submission (submit pl00_intro in Jupyter and UTOL (Assignment 1))
- work on assignment pl01_basics

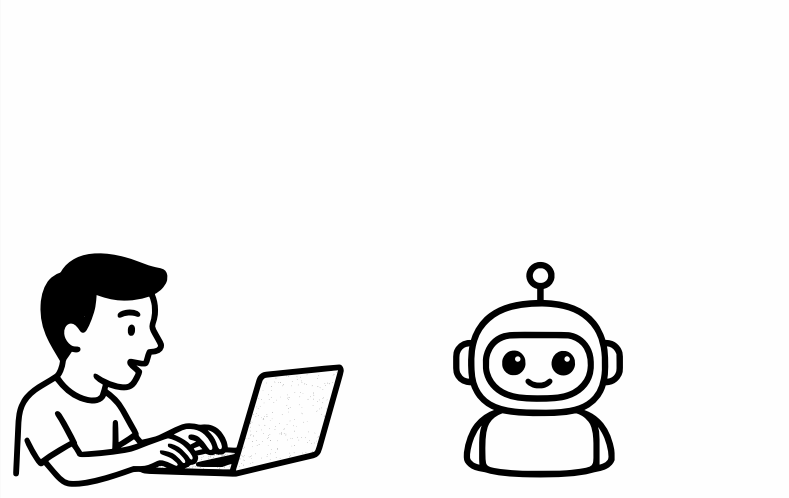
AI Tutor (Hey Tutor!)

- you can ask AI from Jupyter environment
- see the course home page and instructions in the notebook for details
- I encourage you to use AI to *help you learn effectively*, not to delegate your work
- Guidelines are in the home page

Good usage vs. bad usage

- objective of learning is changes in your brain

vs.



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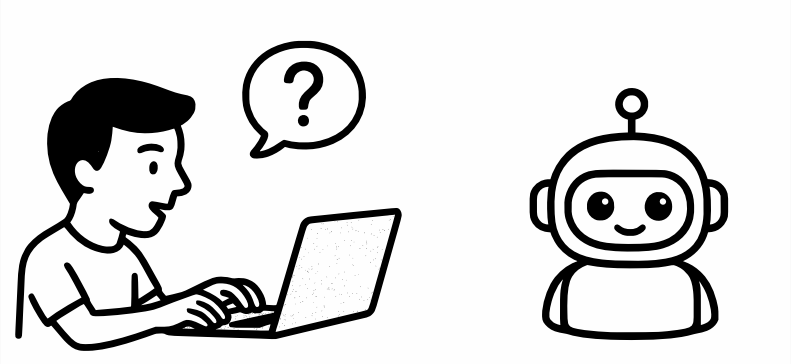

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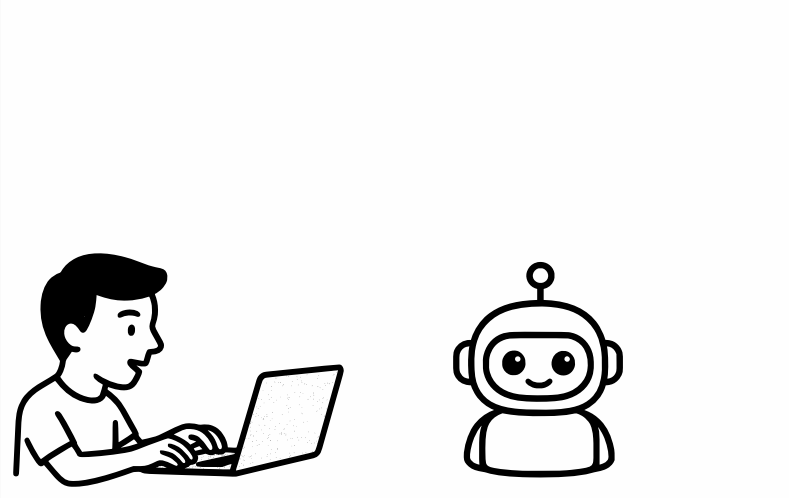

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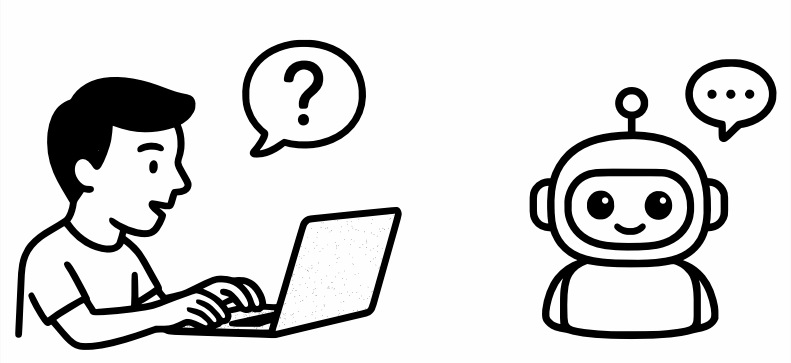

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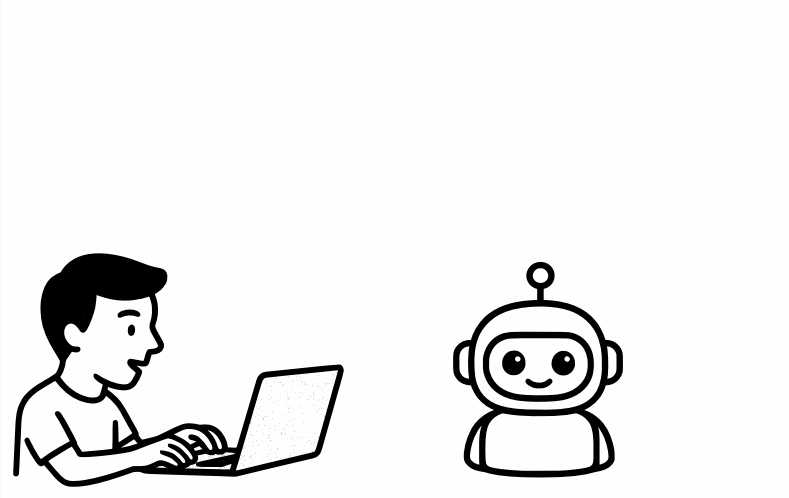

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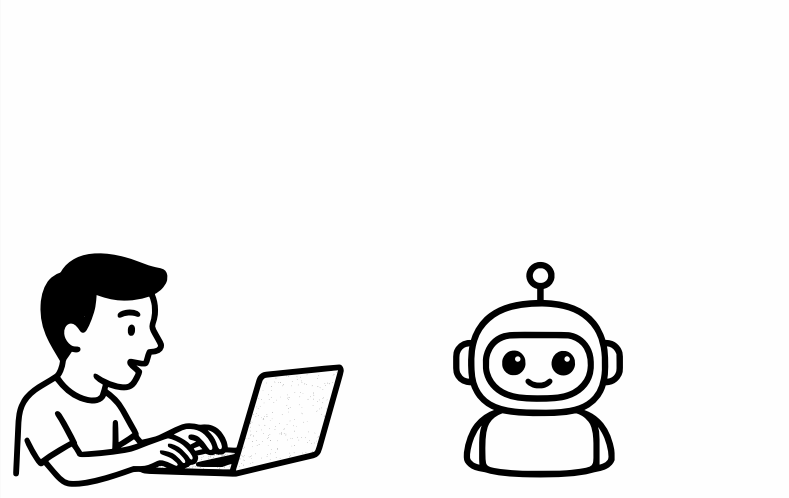

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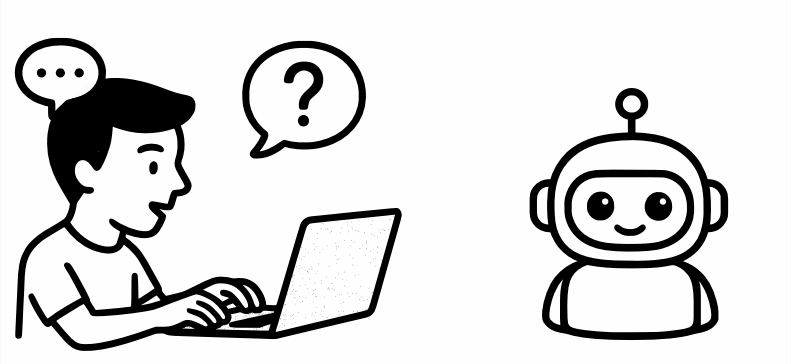

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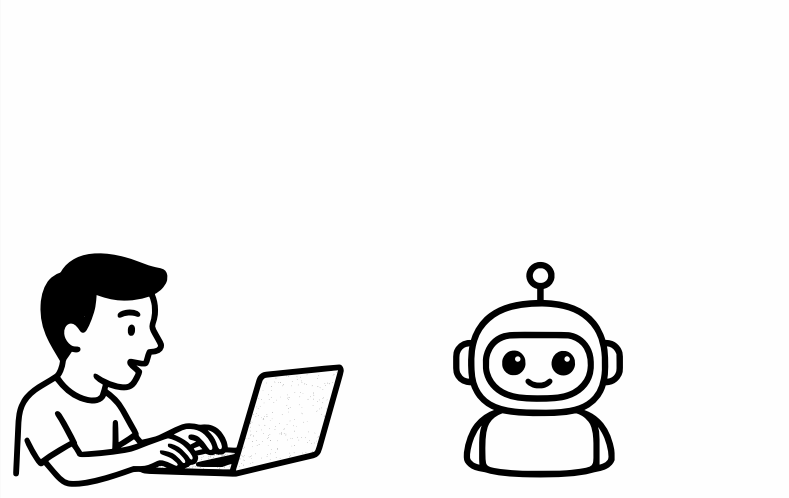

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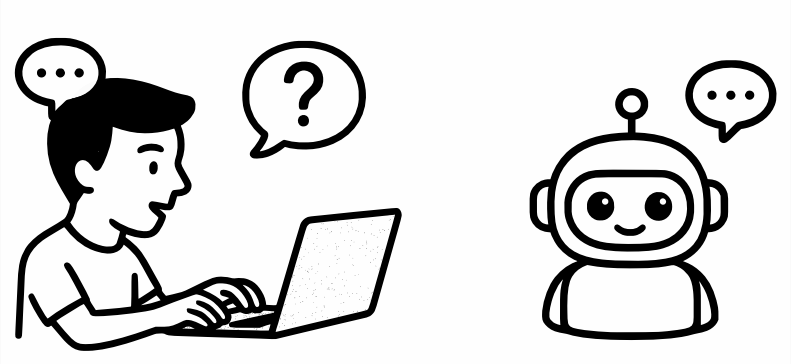

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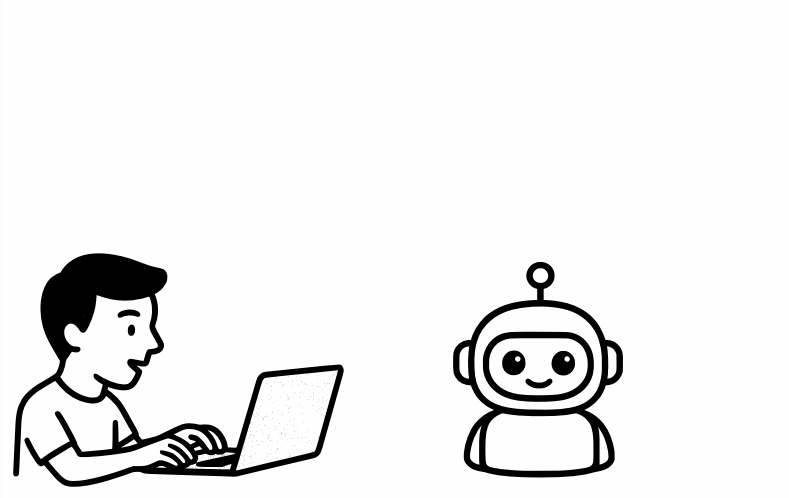

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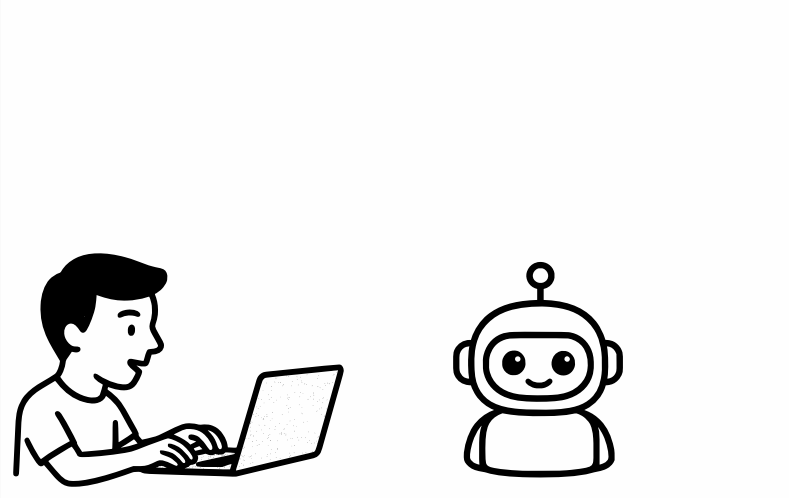

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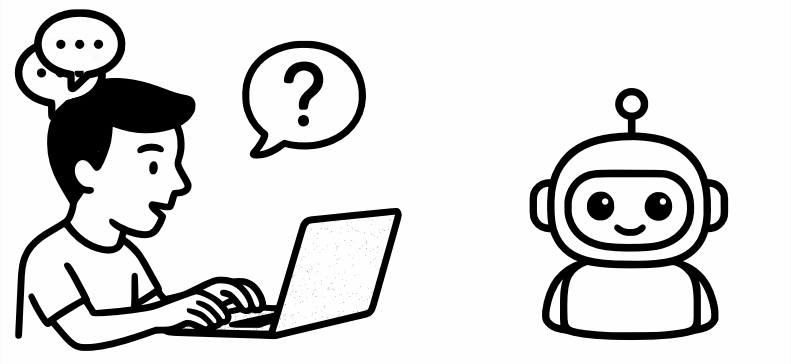

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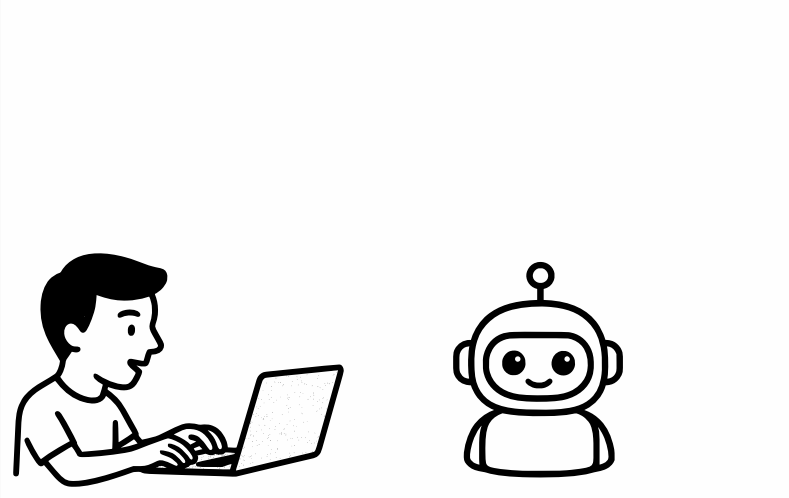

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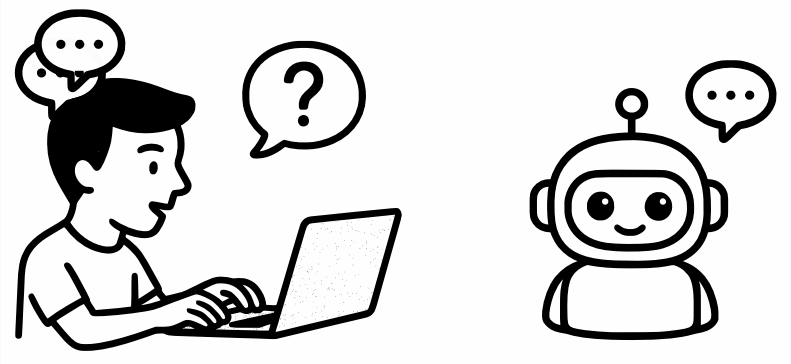

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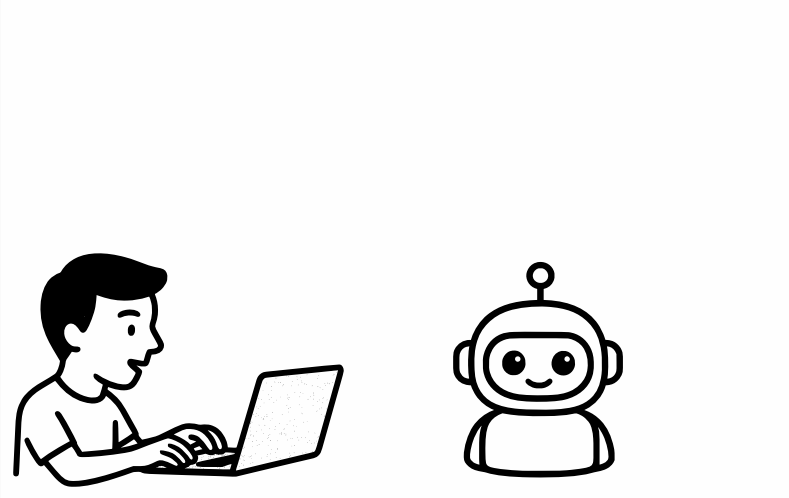

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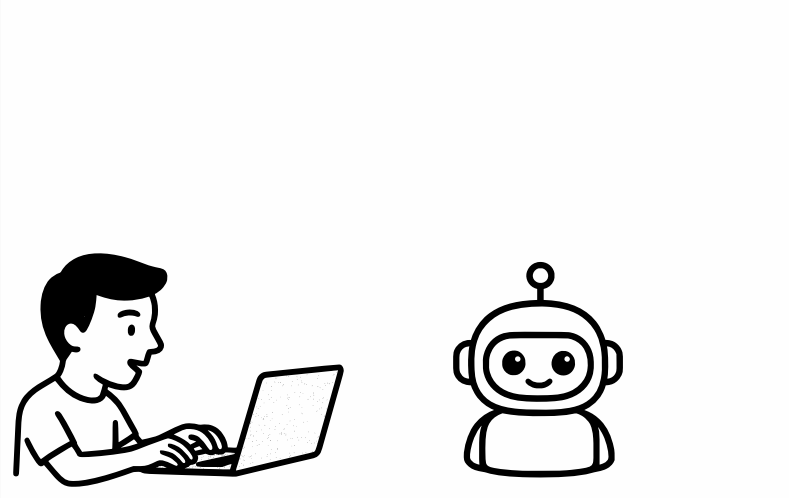

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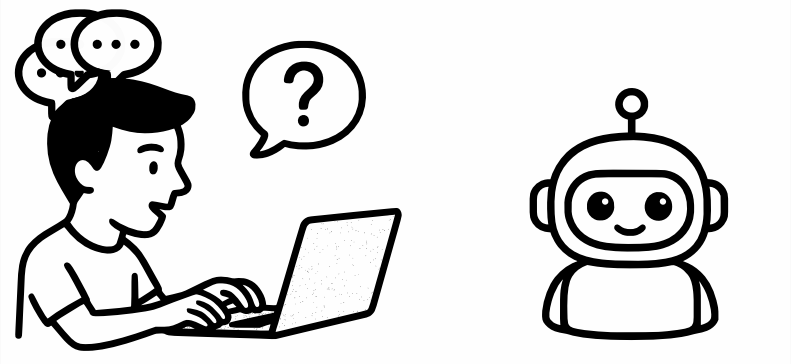

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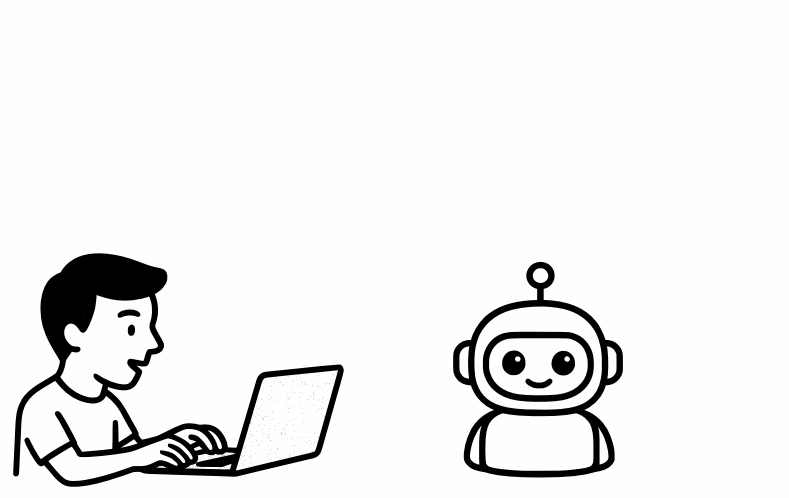

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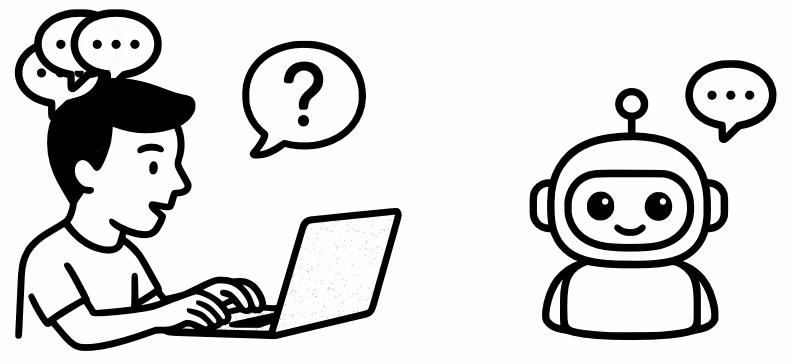

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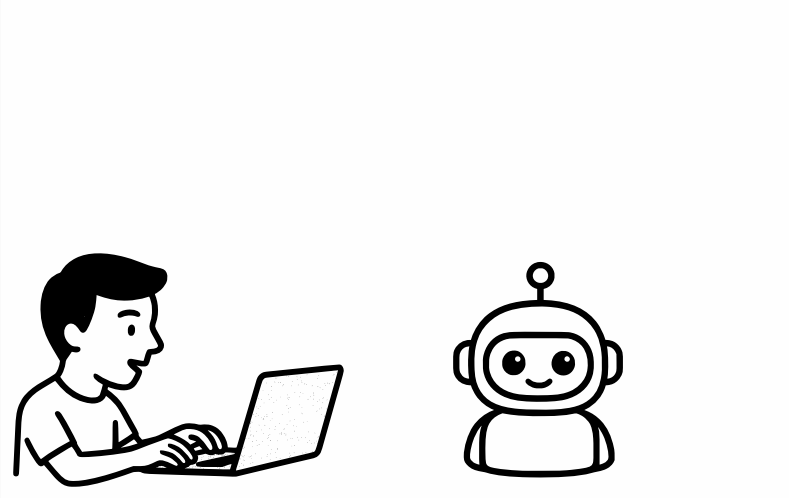

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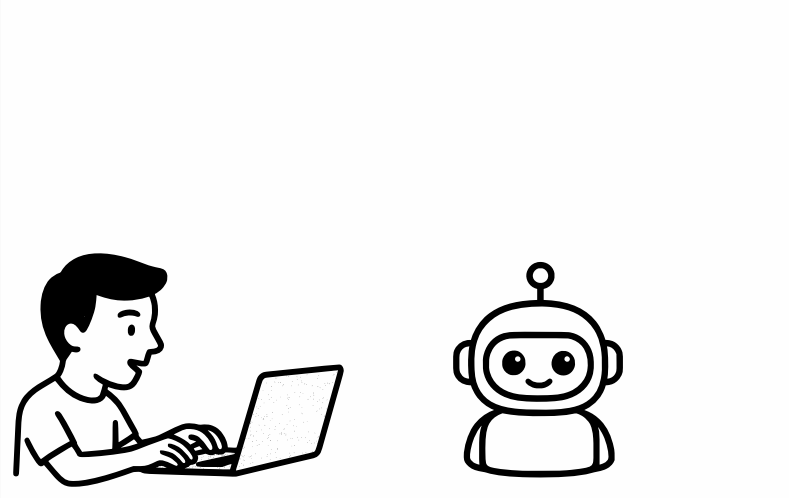

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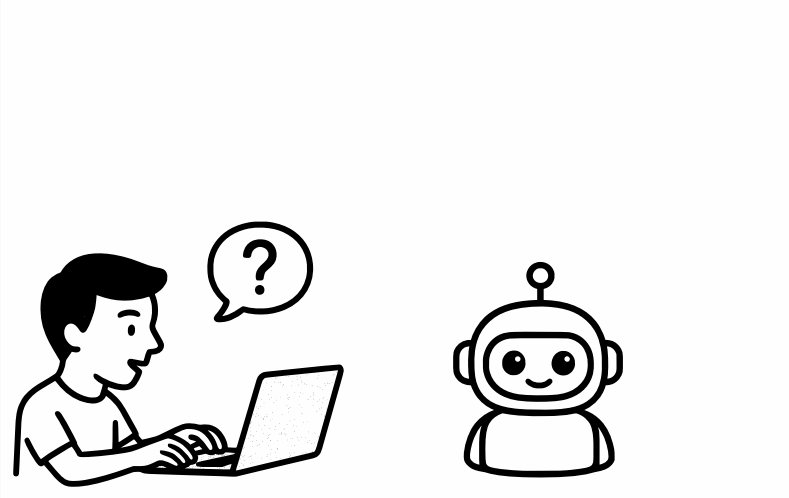

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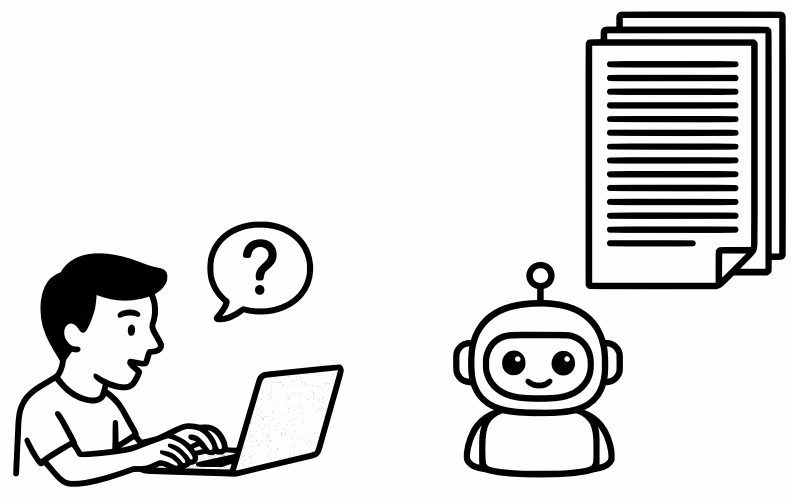

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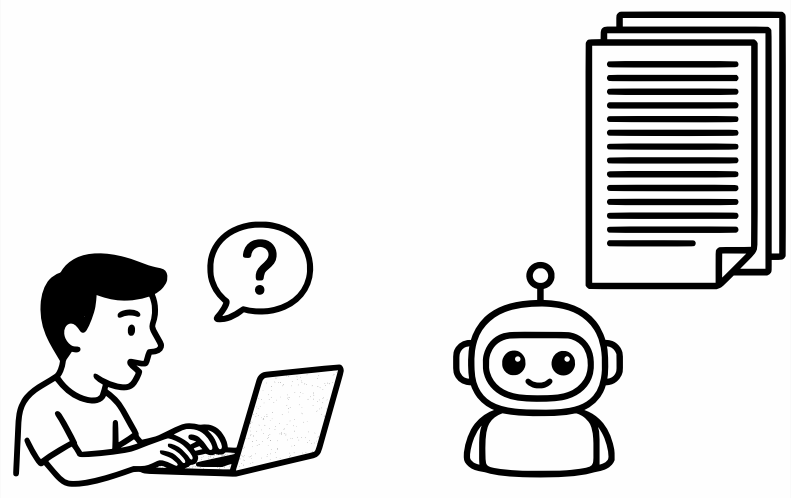
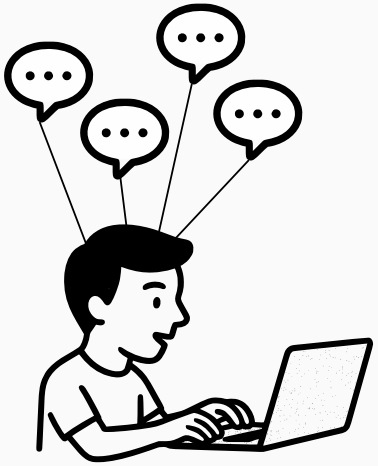


- good : many short interactions you actually absort
- bad : large output you will not read

Fragmented knowledge vs. structured knowledge

- A single interaction with AI only gives you an answer to a narrow question
- Repeated interactions only give you a collection of *disconnected* knowledge and does not stick in memory



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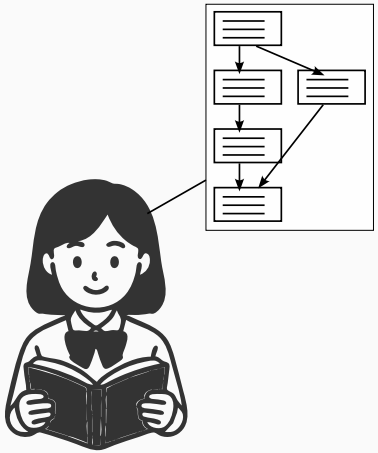
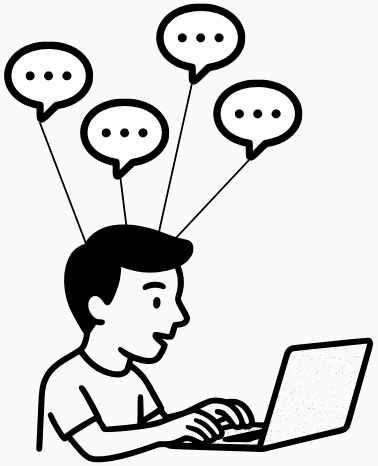
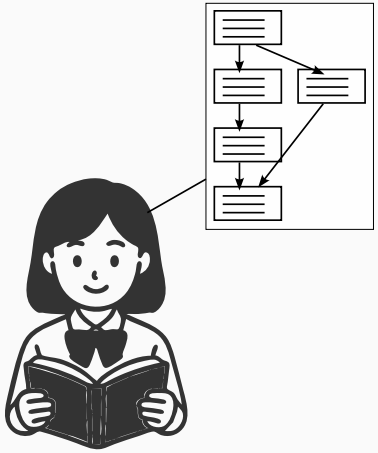


Fragmented knowledge vs. structured knowledge

- Knowledge sticks in your brain when they are *connected* and *structured*
- How to obtain a *connected/structured knowledge*? $\Rightarrow$ (good) textbooks or books on a subject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